THE JOURNAL OF PHYSICAL CHEMISTRY C

Article

pubs.acs.org/JPCC

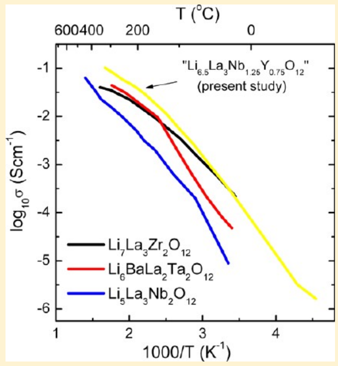
Enhancing Li Ion Conductivity of Garnet-Type $Li_5La_3Nb_2O_{12}$ by Y- and Li-Codoping: Synthesis, Structure, Chemical Stability, and Transport Properties

Sumaletha Narayanan, Farshid Ramezanipour, and Venkataraman Thangadurai*

Department of Chemistry, University of Calgary, 2500 University Drive NW, Calgary, AB, T2N 1N4, Canada

Supporting Information

ABSTRACT: Novel Li-stuffed garnet-like "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" ($0.05 \leq x \leq 0.75$) was prepared via solid-state reaction in air and characterized using *ex situ* and *in situ* powder X-ray diffraction (PXRD), ^{7}Li and ^{27}Al magic angle spinning nuclear magnetic resonance (MAS NMR), scanning electron microscopy (SEM), thermo-gravimetric analysis (TGA), and AC impedance spectroscopy. Rietveld refinement with the PXRD data confirmed the formation of a cubic, garnet-like *Ia*-3*d* structure. ^{7}Li MAS NMR showed a single sharp peak close to 0 ppm as the usual trend for fast Li ion conducting garnets. Among the materials studied, "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" showed a very high bulk ionic conductivity of 2.7×10^{-4} S cm^{-1} at 25 °C, which is the highest value found in garnet-type compounds, and is only reported for $Li_7La_3Zr_2O_{12}$. The *in situ* PXRD measurements revealed structural stability up to 400−600 °C after water treatment as well as chemical compatibility with high voltage Li cathodes $Li_2MMn_3O_8$ (M = Co, Fe). The current work demonstrates that slight Al contamination from the Al_2O_3 crucible, which is commonly observed in this class of materials and was detected by ^{27}Al MAS NMR, does not affect the Li ionic conductivity and chemical stability of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" garnets. It also shows that Li stuffing is critical to improve further the ionic conductivity of parent compound $Li_5La_3Nb_2O_{12}$. Y^{3+} seems to be a very efficient dopant to improve the ionic conductivity of garnets compared to other investigated dopants that include M^{2+} (M = alkaline earth metals), In^{3+}, and Zr^{4+}.



1. INTRODUCTION

Today's world is in a huge demand for high efficiency carbon free energy storage and conversion systems to tackle the global greenhouse emissions. Li ion batteries (LIBs) are a promising candidate for applications in electric vehicles and portable electronics.[1−7] LIBs offer higher volumetric and gravimetric densities compared to other commonly known secondary batteries. The state-of-the-art LIBs are comprised of flammable organic solvents with organic polymer and Li salt electrolytes. All-solid-state LIBs using solid electrolytes are potentially safer compared with the conventional organic liquid membranes. Solid electrolytes exhibiting ionic conductivity in the order of 10^{-4}−10^{-2} S cm^{-1} and high electrochemical stability (up to 5.5 V/Li) have the potential to replace the polymer electrolytes in current LIBs. Various inorganic crystal structure types, including Li silicates (e.g., Li_4SiO_4), Li phosphorus oxynitrides (e.g., $Li_{2.88}PO_{3.73}N_{0.14}$), Na superionic conducting (NASICON) phosphates (e.g., $LiTi_2P_3O_{12}$), A-site deficient perovskite-type oxides (e.g., $La_{(2/3)-x}Li_{3x}\square_{(1/3)-2x}TiO_3$) are being considered as electrolytes for all-solid-state LIBs.[8−11]

Recently reported Li-containing metal-oxides with a garnet-type structure appear to be one of the most promising candidates for all-solid-state LIBs due to their high ionic conductivity and excellent chemical stability in contact with metallic Li anodes and Li cathodes. The original garnet structure has a general formula $A_3B_2Si_3O_{12}$ (where A = Ca, Mg and B = Al, Cr, Fe) and the A, B and Si cations occupy the eight, six, and four coordinated sites, respectively.[12,13] The garnet-type $Li_5La_3M_2O_{12}$ (M = Nb, Ta) can be termed as Li-stuffed garnet-type compounds because of their structural similarity to garnets. The total (bulk + grain-boundary) conductivity of 10^{-6} S cm^{-1} (with activation energy ~0.5 eV) at ambient temperature was reported for $L_5La_3M_2O_{12}$ by Thangadurai et al. in 2003.[14] The Ta phase showed better electrochemical stability in contact with Li and a greater ionic conductivity compared to the Nb compound. It is expected that partial substitution on the La^{3+} site by aliovalent cations can tailor the connectivity of the Li^+ mobility network by controlling the easily accessible Li ion vacancies, and therefore improving the Li ionic conductivity.[15−17] For example, $Li_6La_2BaTa_2O_{12}$ shows an enhanced ionic conductivity in the order of 10^{-5} S cm^{-1} at 22 °C.[16] Studies have shown that substitution at the M-site can result in garnet-like compounds with a conductivity of 10^{-5}−10^{-4} S cm^{-1} at 25 °C.[18−20] Among the known garnets, so far, the cubic $Li_7La_3Zr_2O_{12}$ showed the highest bulk conductivity of 10^{-4} S cm^{-1} which is about 2 orders of magnitude higher than that of the tetragonal phase at 25 °C.[18] Several studies like neutron diffraction, computational

Received: May 15, 2012
Revised: August 31, 2012

   dx.doi.org/10.1021/jp304737x | J. Phys. Chem. C XXXX, XXX, XXX−XXX

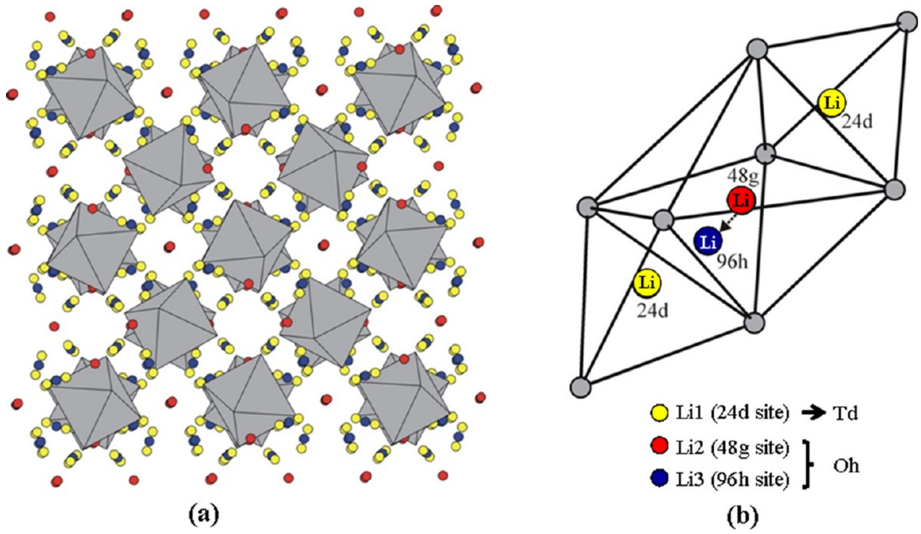


Figure 1. (a) The garnet-type structure of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$". Note the tilting of $Nb(Y)O_6$ octahedra (grey). The red, yellow and blue spheres represent La and two different Li sites. (b) The Li atom environment. The smaller grey spheres are oxygen atoms. The octahedral Li can be at the center (48g site) or shifted (96h site) toward one of the faces shared with the neighboring tetrahedron as shown in Figure 1b. It is important to note that the octahedral position and the two neighboring tetrahedral sites may not be simultaneously occupied, due to electrostatic repulsion, and at least one of the three sites should be empty.[22,28]

simulations, and nuclear magnetic resonance (NMR) spectroscopy are focused to understand the mechanism of Li dynamics in the garnet-like compounds.[13,21−24] There seems to be no report on substitution of Y at the M site in $L_5La_3M_2O_{12}$ to improve the Li^+ conductivity. Even though the Ta analogue shows more stability than that of Nb, we have started with the Nb member as the Nb precursors are comparatively cheaper. Here, we report, for the first time, novel garnet-like "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" and $x = 0.5$ and 0.75 members that exhibit a conductivity of 10^{-4} S cm^{-1} at room temperature which is in the order of the highest reported value for Li stuffed garnets. We also established an important fundamental understanding of the chemical composition-structure-ionic conductivity relationship in "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" together with their thermal stability and chemical stability in water, and high voltage Li cathodes $Li_2MMn_3O_8$ (M = Co, Fe) and carbon anode.

2. EXPERIMENTAL SECTION

2.1. Synthesis of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" ($0.05 \le x \le 0.75$). The conventional solid-state ceramic technique was used for the preparation of compounds having nominal composition "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" using a stoichiometric amount of the precursors of high purity $LiNO_3$ (99%, Alfa Aesar), La_2O_3 (99.99%, Alfa Aesar, preheated at 900 °C for 24 h), Nb_2O_5 (99.5%, Alfa Aesar), and $Y(NO_3)_3$ (99.9%, Alfa Aesar). In order to compensate for the lithium oxide loss during annealing, 10 wt % excess $LiNO_3$ was added. A Pulverisette, Fritsch, Germany ball mill was used to ensure the homogeneous mixing at 200 rpm for 12 h in 2-propanol using zirconium balls before and after heat treatment at 700 °C for 6 h. The resultant powders were pressed into pellets using an isostatic press. Initially, the pellets were sintered at 900 °C for 24 h followed by 1100 °C for 6 h covered with the mother powder in clean alumina crucible.

2.2. Phase and Electrical Conductivity Studies of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" ($0.05 < x < 0.75$). The phase purity was determined by powder X-ray diffraction (PXRD) using a Bruker D8 powder X-ray diffractometer with Cu Kα. The ^{27}Al and ^{7}Li magic angle spinning nuclear magnetic resonance (MAS NMR) (AMS 300, Bruker, at spinning rate 5 kHz) was performed for the powdered samples of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$", and chemical shift values were expressed against solid $Al(NO_3)_3$ and LiCl, respectively. Scanning electron microscopy (SEM) (Philips FEI XL30) was employed to study the morphology. A Solartron SI 1260 impedance and gain-phase analyzer in the frequency range of 0.01 Hz to 1 MHz in air was used to measure the electrical conductivity of the pellets painted with Au electrodes (cured at 600 °C for 1 h). Dry ice was used to control the temperature below room temperature.[25]

2.3. *In Situ* PXRD Chemical Stability of As-Prepared Aged "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" ($x = 0.5$), and with Water, $Li_2MMn_3O_8$ (M = Co, Fe), and Carbon. The chemical stability of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" after treating with H_2O and as-prepared Li garnets and Li cathodes $Li_2MMn_3O_8$ (M = Co, Fe) was tested. For the H_2O stability test, the pellet was crushed into powder and was then treated with H_2O under mechanical stirring for 2 days. The stability toward $Li_2MMn_3O_8$ (M = Co, Fe) and carbon was tested by heating a mixture of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" and the electrode (1:1 wt % ratio) at different temperatures using *in situ* PXRD in air. *In situ* PXRD was performed on the same PXRD machine with high temperature reactor chamber (Anton Paar XRK 900). The samples were heated at a rate of 10 °C/min and then stabilized for 30 min, prior to each measurement in the range 30−900 °C. Thermal stability measurement was conducted using thermogravimetric analysis (TGA) (Mettler Toledo thermal system; TGA/DSC1 HT 1600 °C system) in the range 25−800 °C under N_2 atmosphere. The ramping rate was set to 10 °C/min for both heating and cooling cycles, and two subsequent cycles were performed.

3. RESULTS AND DISCUSSION

3.1. PXRD and Structural Properties. The GSAS program[26] and EXPGUI interface[27] were used for the Rietveld refinements that confirmed the formation of a garnet-type *Ia*-3*d*

Table 1. The Refined Unit Cell Parameters (a (Å)) for "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" Phases

x in "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$"	cell constant, a (Å)
0.05	12.8169(6)
0.10	12.8216(4)
0.20	12.8306(7)
0.25	12.8582(5)
0.50	12.9136(4)
0.75	12.9488(11)

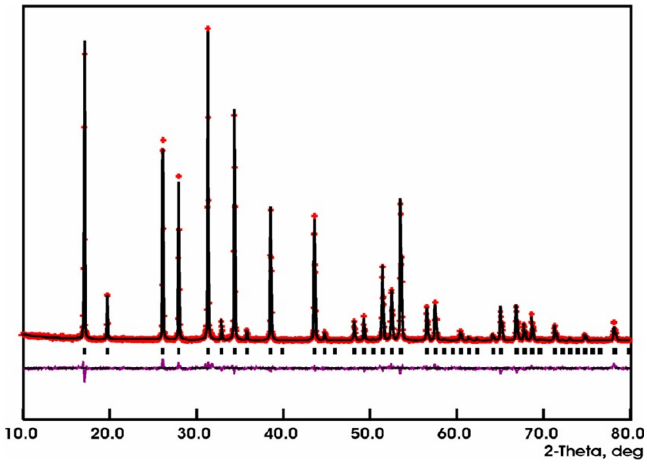


Figure 2. The powder X-ray Rietveld refinement profile for "$Li_{5.2}La_3Nb_{1.9}Y_{0.1}O_{12}$".

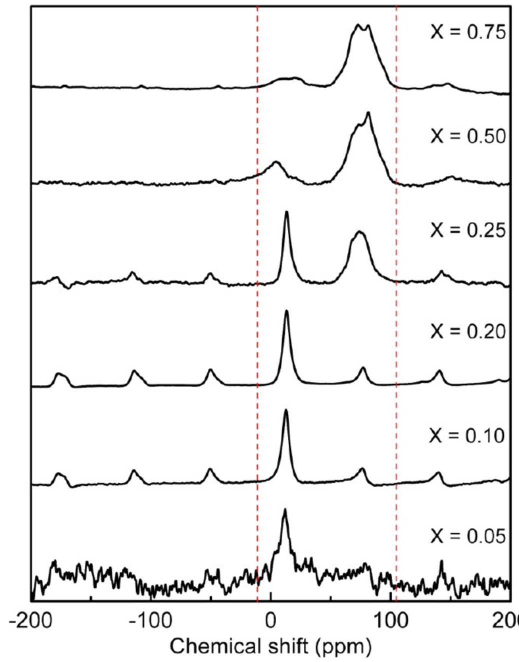


Figure 3. ^{27}Al MAS NMR of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" measured at a spinning frequency of 5 kHz. The chemical shift values are expressed with respect to $Al(NO_3)_3$.

structure for all phases. The structure contains Nb(Y) atoms coordinated octahedrally by oxygen. There is no direct corner- or edge-sharing between the $Nb(Y)O_6$ octahedra. They are also tilted to the left or right relative to the unit cell axis direction, such that the tilting of each octahedron is opposite to its nearest neighbor octahedra (Figure 1). The refined cell parameters for "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" phases (Table 1) range from 12.8169(6) to 12.9488(11) Å, close to the reported cell constants for $Li_5La_3M_2O_{12}$ (M = Nb, Ta)[22] (12.80654(11) Å) and $Li_7La_3Zr_2O_{12}$ (13.0035 Å).[28] The obtained cell parameters are consistent with the doping level and the larger ionic radius of Y^{3+} (0.90 Å) compared to those of Nb^{5+} (0.64 Å), Ta^{5+} (0.64 Å), and Zr^{4+} (0.72 Å).[29] Among the synthesized materials, the x = 0.2 and 0.75 phases show the presence of small peaks belonging to an unidentified second phase. Note that the parent compound, $Li_5La_3Nb_2O_{12}$,[22] has also been reported to be accompanied by impurity. The impurity peaks vanish in x = 0.05, 0.1, 0.25, and 0.5 phases, and only very small humps, barely distinguishable from the background, are observed.

Figure 2 shows a typical Rietveld refinement profile for "$Li_{5.2}La_3Nb_{1.9}Y_{0.1}O_{12}$", and Table 2 presents the refined atomic parameters for the nearly pure phase "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$". The refinement profiles and the refined atomic parameters of all other members are provided in the Supporting Information (Figures S1−S5 and Tables S1−S5). While the Li positions cannot be determined by PXRD, some insight can be obtained from the available neutron diffraction studies on Li positions in garnets. In $Li_5La_3M_2O_{12}$ (M = Nb, Ta),[22] lithium is distributed over three positions: tetrahedral site (24d), center of the octahedra (48g site), and shifted from the center of octahedra (96h site) as shown in Figure 1b. The displacement of Li^+ cations from the center of octahedra is justified in terms of reduction in electrostatic repulsion and prevention of very short Li−Li distances that can occur between a Li atom on the tetrahedral site and its neighboring octahedral Li.[22,30] The high Li^+ mobility is described on the basis of short Li^+−Li^+ distances in the edge-sharing LiO_6 octahedra that facilitate the Li^+ conduction.[22] A somewhat different picture of Li positions is presented for $Li_7La_3Zr_2O_{12}$,[28] where the octahedral Li atoms are all shifted from the center of the octahedra. However, regardless of the differences between details of these structural models, it is evident that higher Li content per formula unit

Table 2. The Powder X-ray Rietveld Refinement Results for "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$"($a$ = 12.9136(4) Å; R_p = 9.64%; R_{wp} = 14.6%)

atom	Wyckoff-site	x	y	z	occupancy	U_{iso}
La	24c	1/8	0	1/4	1	0.028(6)
Nb	16a	0	0	0	0.75	0.027(6)
Y	16a	0	0	0	0.25	0.027(6)
Li1[a]	24d	1/4	7/8	0	0.674	0.030
Li2[a]	48g	1/8	0.6773	0.5727	0.220	0.030
Li3[a]	96h	0.0937	0.6888	0.5817	0.218	0.030
O	96h	0.2883(8)	0.1009(8)	0.1969(8)	1	0.029(7)

[a]The Li position and occupancies are taken from ref 30.

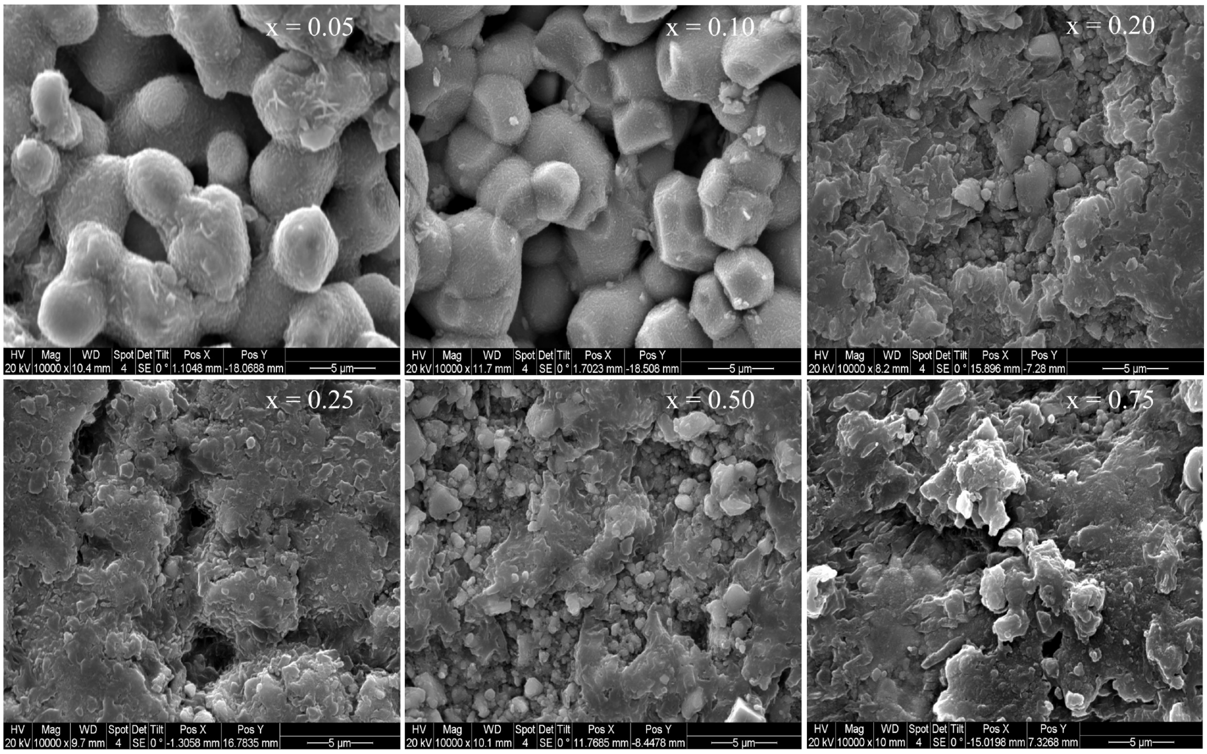


Figure 4. Typical SEM images of as-prepared "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" in air.

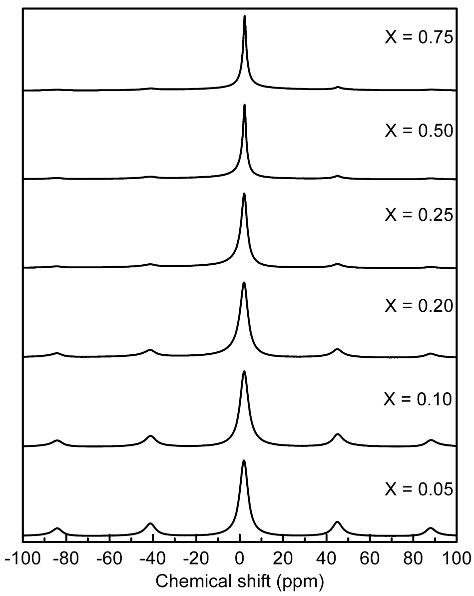


Figure 5. ^7Li MAS NMR of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" measured at a spinning frequency of 5 kHz. The chemical shift values are expressed with respect to solid LiCl.

comes with a greater octahedral site occupation,[22,28] as also shown for $Li_{5+x}Ba_xLa_{3-x}Ta_2O_{12}$ (x = 0−1.6).[30] In addition, a high concentration of Li on the octahedral sites appears to result in greater Li^+ conductivity.[15,22,28,30] In $Li_3Nd_3Te_2O_{12}$,[31] where Li is located exclusively on the tetrahedral site, a low Li^+ conduction was reported. If our materials follow the same trend, a significant Li occupation on the octahedral sites could be expected, considering their high lithium ion conduction, as discussed in the section 3.3 of this paper.

3.2. ^{27}Al and ^7Li MAS NMR and Morphology.

As the preparation process involved the use of alumina crucibles, in order to check the possibility of Al impurity,[20b,24] the ^{27}Al NMR is also recorded and the data is provided in Figure 3. Surprisingly, a chemical shift value (δ) close to 15 ppm was observed, which indicates an octahedrally coordinated Al and one which is greater than 70 ppm, indicative of a tetrahedrally coordinated Al. It is known that α-Al_2O_3 shows an NMR peak with δ in the range 13−17 ppm, and θ-Al_2O_3 shows two peaks centered at 11(1) and 80(1) ppm, as it has an equal number of octahedral and tetrahedral sites.[20b,27] The possible peaks which can be seen due to the transformation of γ-AlOOH at higher temperature to γ-Al_2O_3 are at 12 and 70 ppm and for α-Al_2O_3, between 5 and 9 ppm.[32] The presence of a peak at ~80 ppm was also reported recently for Al-doped $Li_7La_3Zr_2O_{12}$.[24] At lower concentrations of Y in "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$", the peak close to 13 ppm is visible, while at higher concentrations of Y only the peak close to 80 ppm is seen. The concentration of Al in garnet was considered to be rather negligible compared to the doping level.

Figure 4 shows the SEM images of the as-prepared garnets "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" in air. The samples were cut into small discs using the diamond saw for the measurements. At lower Y dopant concentrations, the grains are clearly visible, and as the Y doping increases, the crystallite to crystallite contact increases and the grains are not distinguishable. This could be due to the fact that Y doping increases the sinterability of the garnets. The ^7Li MAS NMR measured using the reference LiCl is shown in Figure 5. A single peak close to 0 ppm was obtained in all cases which is characteristic of the garnet-like compounds reported elsewhere.[17,19,20a,33,34] Li ion mobility is directly proportional to the narrowness of the peak, and therefore, a narrower peak indicates a higher Li ion mobility.[33]

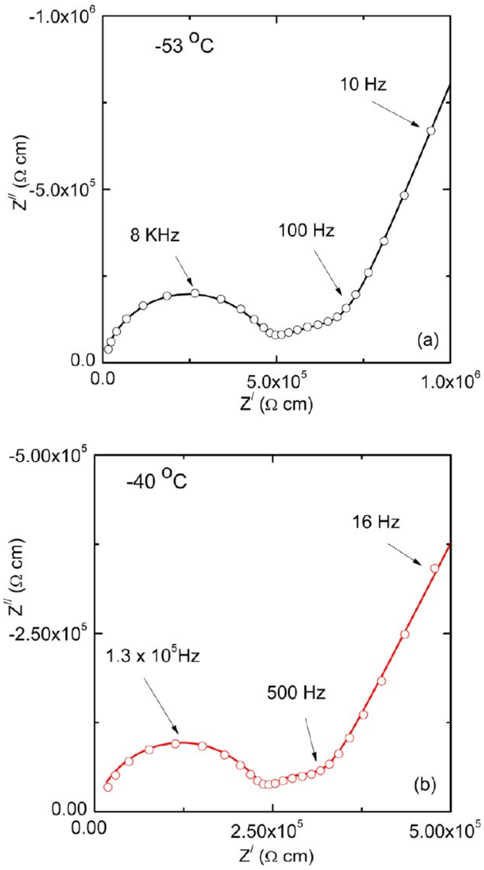


Figure 6. Typical AC-impedance plots of "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" measured at (a) −53 °C and (b) −40 °C. Solid lines show fits using the equivalent circuit containing individual resistances and constant phase elements (CPEs) representing the electrical bulk, grain-boundary, and electrode responses. The χ^2 values obtained were very low, $\sim 10^{-3}$, showing good fits.

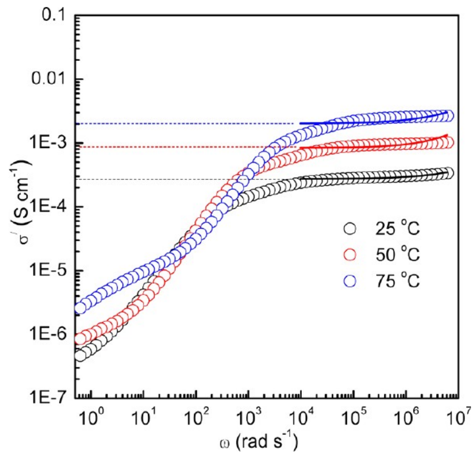


Figure 7. Impedance analyses of "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" measured at the temperatures indicated. Solid lines represent a rough analysis of the data using a power law expression (see eq 2 and Table 4). The frequency range used for power law fit is 10^4–10^6 Hz.

3.3. Electrical Properties. Figure 6 shows typical AC-impedance plots of "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" measured below the room temperature (a and b represent −53 and −40 °C, respectively). Solid lines show fits using the equivalent circuit containing individual resistances and constant phase elements, representing the electrical bulk, grain-boundary, and electrode responses. The equation used to calculate the capacitance (C) value is given below.[35]

$$C = R^{((1-n)/n)}Q^{(1/n)} \tag{1}$$

where R is the resistance, Q is the constant phase element, and n is a parameter that has a value close to 1. The capacitance C for the high-frequency semicircle is found to be in the order of 10^{-11} F, indicating bulk contribution, and the semicircle appearing at intermediate frequencies with a capacitance in the order of 10^{-9} F (Table 3) indicates grain-boundary contribution.[14,15,36] The χ^2 values obtained were very low, $\sim 10^{-3}$, indicating reasonably good fits.

Figure 7 shows the impedance analysis of "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$", measured at 25, 50, and 75 °C. Solid lines represent a rough analysis of the data using a power law, eq 2, and Table 4 summarizes the parameters derived from the fitting. A few examples are also provided for comparison.[23,37,38] A plateau is observed at higher frequencies (10^4–10^6 Hz), indicating the frequency independent nature of the conductivity and the so-called dispersive AC-conductivity region.[39,40] At lower frequency region, the polarization effects due to ion blocking electrodes arise, and frequency dependency of the conductivity is clearly visible. The dependence of $\sigma(\omega)$ on frequency can be expressed by the following power law expression:[39,40]

$$\sigma(\omega) = \sigma(0) + A\omega^p \tag{2}$$

where $\sigma(0)$ represents the DC-conductivity and p the power law exponent (shows values between 0.7 and 0.8). The p values for ionic conductors are generally between 1 and 0.5, indicating the long-range and tortuous pathways, respectively.[40,41] Experimental limitations in the access of dispersive region should be taken into account, and hence, the power law exponents have to be considered as rough estimations. However, the values obtained are in agreement with results reported for structurally disordered materials, like garnet-type structured Li stuffed oxides,[13,21,42] and other solid state Li ion electrolytes.[37,38]

An Arrhenius plot of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" is shown in Figure 8. The electrical conductivity increases with increasing temperature and follows semiconducting behavior. A highest conductivity of 10^{-4} S cm^{-1} was observed for both "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" and "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$" at 24 °C. The activation energy calculated for "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" at 23−325 °C is in the range 0.36−0.48 eV; as commonly observed for garnet-type oxides, the activation energy decreased with increase in Li content in the garnet.[14−21] The diffusivity

Table 3. Fitting Results of the Analysis of the AC-Impedance Data (Figure 6) by Using Suitable Equivalent Circuit Elements

T (°C)	R_b (Ω)	CPE_b (F)	C_b (F)	R_{gb} (Ω)	CPE_{gb} (F)	C_{gb} (F)	χ^2
−53	1.66×10^5	7.59×10^{-11}	1.42×10^{-11}	6.93×10^5	3.22×10^{-8}	3.99×10^{-9}	0.0012
−40	8.26×10^4	1.23×10^{-10}	1.40×10^{-11}	2.42×10^4	1.59×10^{-8}	4.94×10^{-9}	0.0011

dx.doi.org/10.1021/jp304737x | J. Phys. Chem. C XXXX, XXX, XXX−XXX

Table 4. Fitting Parameters Obtained from the Frequency-Independent Plateaus (10^4–10^6 Hz) of the Impedance Spectra for "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" Provided in Figure 7

compound	T (°C)	σ_0 (S cm^{-1})	p	reference
$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$	25	2.7×10^{-4}	0.65	present work
	50	8.3×10^{-4}	0.80	
	75	1.2×10^{-3}	0.68	
$Li_7La_3Zr_2O_{12}$	27	1.2×10^{-6}	0.77 (< −135 °C)	23
$Li_{0.18}La_{0.61}TiO_3$	30	1.0×10^{-4}	0.60	36
Li_4SiO_4	123	4.0×10^{-8}	0.68	35

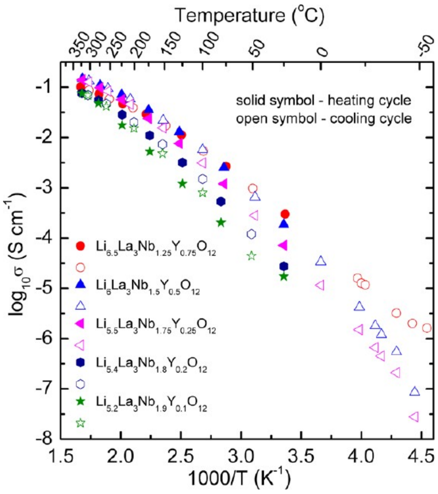


Figure 8. Arrhenius plots of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" ($x$ = 0.1–0.75) at different temperatures ranging between −53 and 330 °C. The solid and open symbols represent heating and cooling cycle data, respectively.

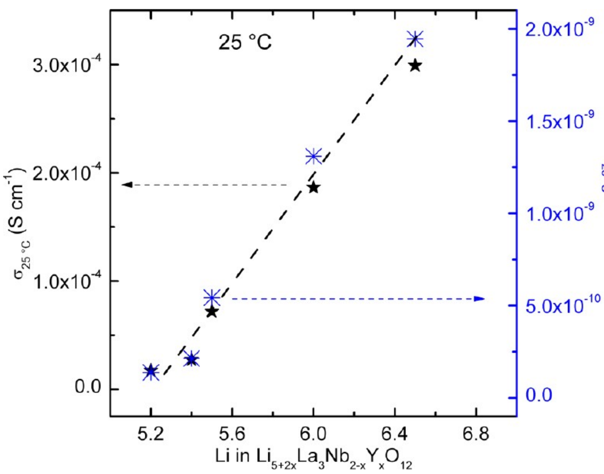


Figure 9. The change in conductivity and diffusivity of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" ($x$ = 0.1–0.75) with change in Li concentration in the formula unit at 25 °C. The lines passing through the data points are guides to the eye.

(D) of Li ions can be calculated from the electrical conductivity using the equation

$$\sigma(0) = \frac{q^2 N}{k_B T} \cdot D \tag{3}$$

where q is the elementary charge (1.6×10^{-19} C), k_B Boltzmann's constant (1.38×10^{-23} J K^{-1}), and N the concentration of charge carriers calculated according to $N = n/V$, where n is the number of Li ions per unit cell and V the volume of the unit cell calculated from the PXRD analysis. The calculated diffusivity (D) value was found to fall in the range 10^{-10}–10^{-8} cm^2 s^{-1} at 25–100 °C. Previous studies have revealed that the diffusivity values (self-diffusion coefficient) calculated from NMR are in good agreement with those of the impedance analysis.[20b,23] A comparison of changes in conductivity (σ) and diffusivity (D) of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" at 25 °C with change in Li concentration in the formula unit is provided in Figure 9. As expected, both the σ and D increase with Li concentration.[20] An increase in conductivity was observed with the increase in concentration of the Li ion in "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$". The availability of more mobile ions contributes toward the total conductivity. The end members of the present study, especially "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$", show conductivities as high as those of the outstanding garnet-like compound, $Li_7La_3Zr_2O_{12}$, and a comparison of electrical conductivities of some important garnet-like materials is given in Figure 10.[14,15,18b,19,20b,42]

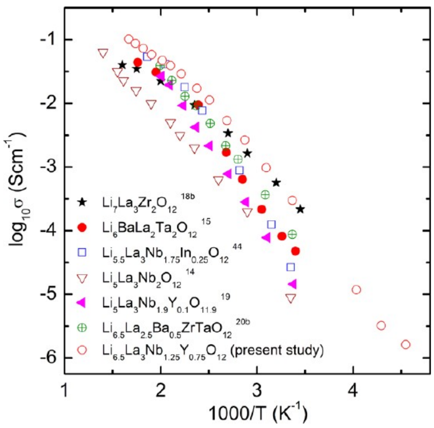


Figure 10. Comparison of Li ion conductivity of "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" with that of other garnets in the literature.[14,15,18b,19,20b,42]

3.4. Thermal Stability of As-Prepared Aged and Water-Treated Garnets, and Chemical Compatibility with Li Cathodes. The garnet-type structure was retained after temperature treatments between 30 and 800 °C for both members, as confirmed by *in situ* PXRD (Figure 11). The thermal stability of as-prepared aged "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" garnets was studied using TGA, and data is shown in Figure 12. In all cases, the weight loss started at ~250 °C and ended at ~600 °C for the first run, which may be due to uptake of water and Li_2CO_3 formation upon exposure to ambient air, as shown in Figure S6 (Supporting Information). The weight loss in this region may be due to water in the crystalline lattice.[34] Very interestingly, the weight loss decreased (from ~8 to 2%) with an increase in Li content in "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$". Only single

dx.doi.org/10.1021/jp304737x | J. Phys. Chem. C XXXX, XXX, XXX−XXX

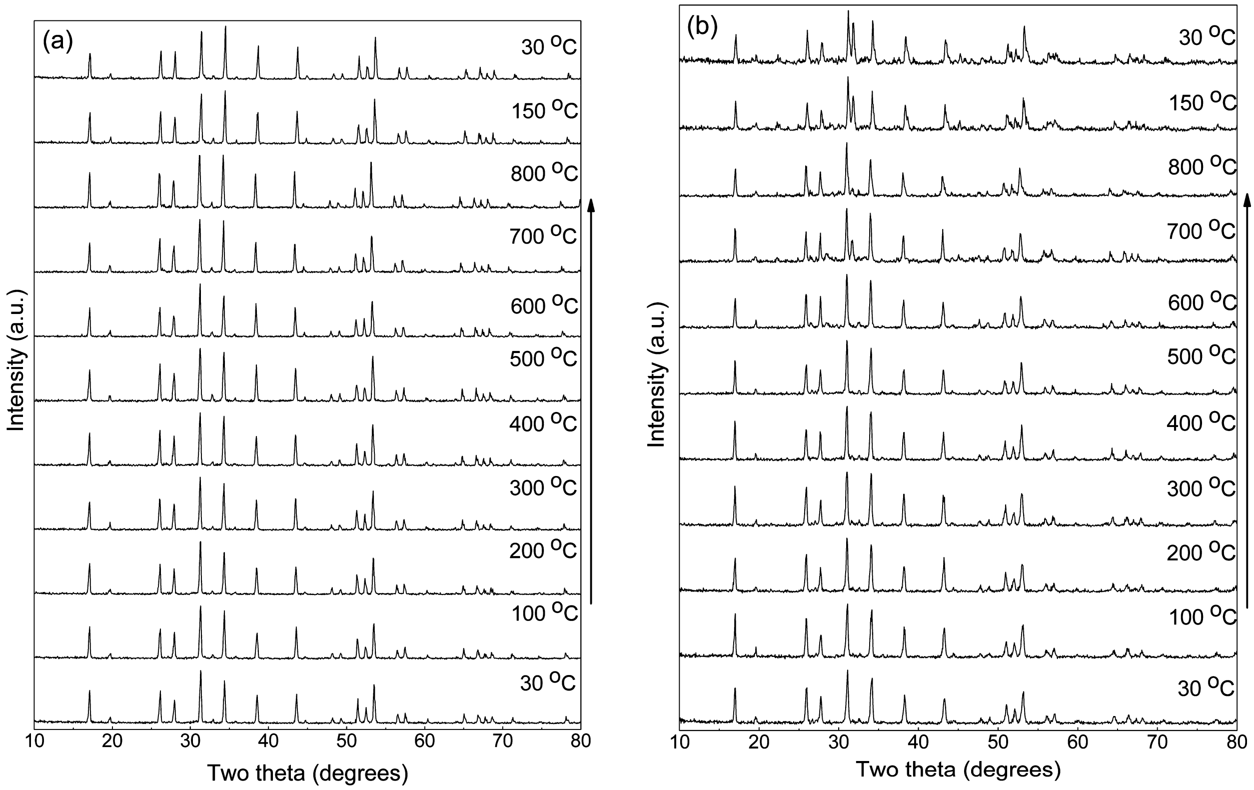


Figure 11. *In situ* PXRD of as-prepared aged "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$": (a) x = 0.05, (b) x = 0.5 in air. The arrow indicates the sequence of measurement. The samples were equilibrated for 30 min prior to each measurement.

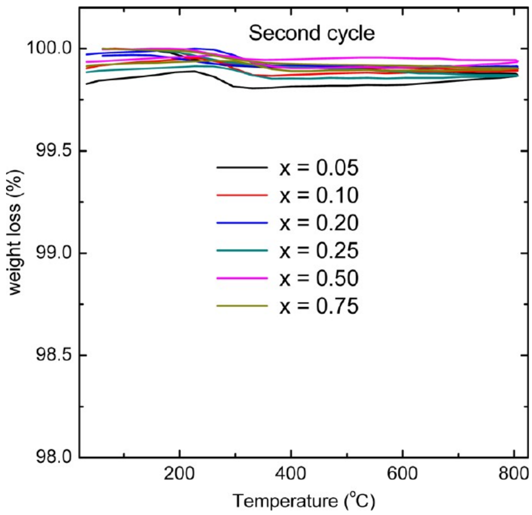


Figure 12. Thermogravimetric analysis (TGA) of as-prepared aged "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" ($x$ = 0.05–0.75) garnets (second cycle) in air.

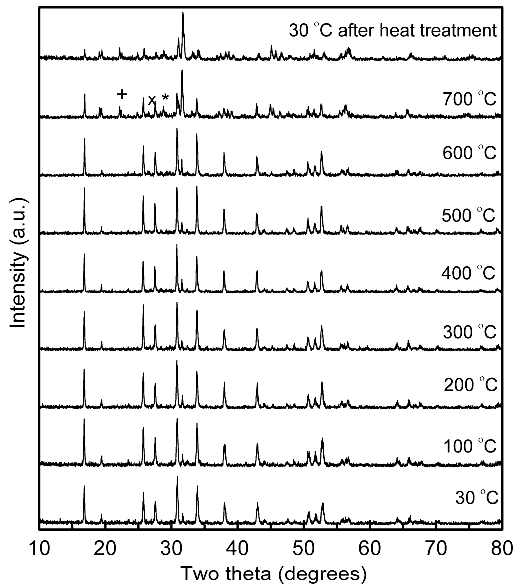


Figure 13. *In situ* PXRD showing the stability of water-treated "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" at different temperatures. The decomposed product above 600 °C could be identified as $LiLaNb_2O_7$ (ICSD No. 00-027-1249 (*) and 01-081-1193 (×)) and $LiLa_2NbO_6$ (ICSD No. 00-040-0895 (+)).

step weight loss is visible for the "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" phases with $x \leq 0.25$. Exceptionally, for the high lithium containing garnets, "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$" and "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$", an additional weight loss of 0.5 and 1%, respectively, was observed after treatment at 600 °C. The second run was carried out to show that the material is stable upon heat treatment of 800 °C. As expected, no weight loss was observed in the second run (Figure 12). Rather interestingly, irrespective of significant weight loss, the garnet-type structure was retained in the temperature range 30–800 °C for x = 0.05 and 0.5 members, as confirmed by *in situ* PXRD (Figure 11).

The possibility of exposure to air can cause water absorption.[34] Hence, "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" was treated with water and an *in situ* PXRD was recorded at varying temperatures to understand the stability of water-treated samples; the result is shown in Figure 13. The garnet-like electrolyte material shows the same cubic structure even after

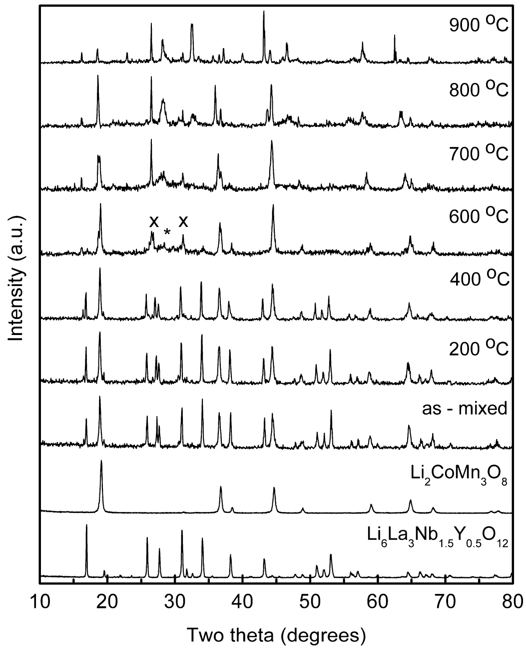


Figure 14. *In situ* PXRD showing the stability of "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$" in the presence of high voltage cathode $Li_2CoMn_3O_8$ at different temperatures. In both cases, the electrolyte showed a high stability up to 400 °C and decayed after further increase in temperature. The impurities are identified as $LiLaNb_2O_7$ (ICSD No. 00-027-1249 (*) and 01-081-1193 (×)) and $LiLa_2NbO_6$ (ICSD No. 00-040-0895 (+)).

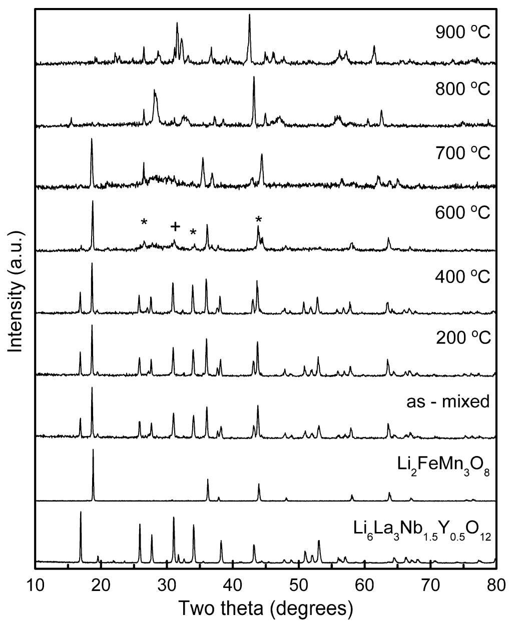


Figure 15. *In situ* PXRD showing the stability of "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$" in the presence of high voltage cathodes $Li_2FeMn_3O_8$ at different temperatures. In both cases, the electrolyte showed a high stability up to 400 °C and decayed after further increase in temperature. The impurities are identified as $LiLaNb_2O_7$ (ICSD No. 00-027-1249 (*) and 01-081-1193 (×)) and $LiLa_2NbO_6$ (ICSD No. 00-040-0895 (+)).

the water-treatment. A few impurity peaks due to $LiLaNb_2O_7$ (ICSD No. 00-027-1249 and 01-081-1193) and $LiLa_2NbO_6$ (ICSD No. 00-040-0895) appeared at 700 °C and are marked in Figure 13 using specific symbols. It is important to note that water-treated Li garnets are stable up to 600 °C and the present

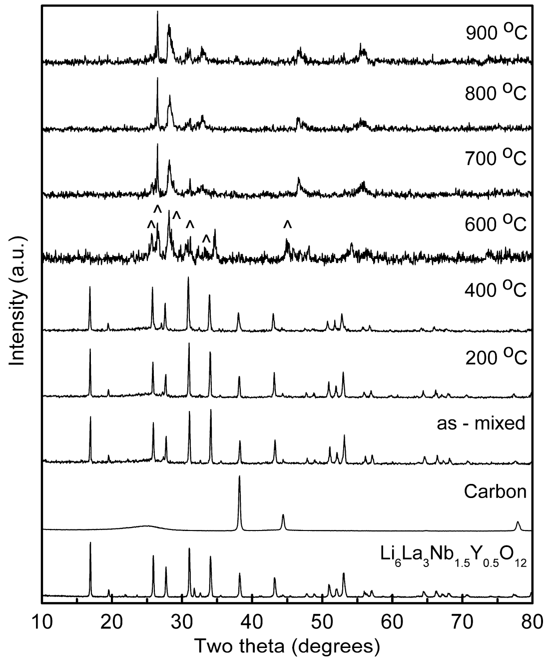


Figure 16. *In situ* powder X-ray diffraction pattern showing the stability of "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$" in the presence of carbon anode at different temperatures. The garnet electrolyte showed a high stability up to 400 °C. Impurity peaks due to the formation of carbonate, $Li_{0.52}La_2O_{2.52}(CO_3)_{0.74}$ (ICSD No. 01-084-1965), are shown by a caret (∧).

result is akin to our earlier claim: a complete decomposition of the water treated (H^+-exchanged) garnet phase was noticed on heating at 700 °C.[43]

The chemical stability and compatibility of the highest conducting phases, "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" and "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$", toward different Li battery components were tested. Figures 14 and 15 show the chemical stability of "$Li_6La_3Nb_{1.5}Y_{0.5}O_{12}$" in the presence of high voltage cathode materials $Li_2CoMn_3O_8$ and $Li_2FeMn_3O_8$, respectively, analyzed using *in situ* PXRD. In both cases, a high stability up to 400 °C was observed for the garnet, and after that, the spinel peaks are found to be the major peaks. Perovskite peaks due to $LiLaNb_2O_7$ (ICSD No. 00-027-1249 and 01-081-1193) appeared above 400 °C and are also marked in Figures 14 and 15. The *in situ* PXRD results indicating the stability of the garnet electrolyte toward the anode material (carbon) are shown in Figure 16. The electrolyte was inert to carbon up to 400 °C and started reacting above this temperature. Impurity peaks due to the formation of carbonate, $Li_{0.52}La_2O_{2.52}(CO_3)_{0.74}$ (ICSD No. 01-084-1965) are marked in the pattern. The higher stabilities under the above-mentioned conditions make these compounds of considerable interest for future studies on solid state electrolytes.

4. CONCLUSIONS

The garnet-like "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" is found to be a promising candidate for future all-solid-state energy storage systems. The PXRD refinements show a cubic crystal structure with space group *Ia*-3*d*. The electrical conductivity measurements show very high σ for the higher lithium stuffed members. "$Li_{6.5}La_3Nb_{1.25}Y_{0.75}O_{12}$" showed a conductivity of 10^{-4} S cm^{-1} at 24 °C which is the same order of magnitude as that of the best garnet-like compound known so far, $Li_7La_3Zr_2O_{12}$. The diffusivity was found to be increasing with increase in the Li content in the formula unit. "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" showed a higher structural stability in water compared to the parent compound. Also, the chemical stability of "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" toward high voltage cathode spinels ($Li_2MMn_3O_8$, M = Co, Fe) is very promising, and a high stability up to 400 °C was observed. The advantages such as very good Li^+ conductivity and high chemical stability make "$Li_{5+2x}La_3Nb_{2-x}Y_xO_{12}$" a strong candidate for future battery research. However, chemical stability of Li-stuffed garnets under humidity/aqueous condition has yet to be studied in detail. The lack of chemical stability under ambient condition suggests that care should be taken to handle the garnet-type solid Li ion electrolytes.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

Tables showing additional powder X-ray Rietveld refinement results and figures showing Rietveld refinement profiles and thermogravimetric analysis. This material is available free of charge via the Internet at http://pubs.acs.org.

■ AUTHOR INFORMATION

Corresponding Author

*Phone: 1 403 210 8649. Fax: 1 403 210 9364. E-mail: vthangad@ucalgary.ca.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

One of the authors (V.T.) would like to thank the AUTO21 Network of Centers of Excellence, Canada's national automotive research and development program for the funding.

■ REFERENCES

(1) Scrosati, B.; Hassoun, J.; Sun, Y.-K. *Energy Environ. Sci.* **2011**, *4*, 3287−3295.
(2) Goodenough, J. B.; Kim, Y. *Chem. Mater.* **2010**, *22*, 587−603.
(3) Ruzmetov, D.; Oleshko, V. P.; Haney, P. M.; Lezec, H. J.; Karki, K.; Baloch, K. H.; Agrawal, A. K.; Davydov, A. V.; Krylyuk, S.; Liu, Y.; et al. *Nano Lett.* **2012**, *12*, 505−511.
(4) Manthiram, A. *J. Phys. Chem. Lett.* **2011**, *2*, 176−184.
(5) Scrosati, B.; Garche, J. *J. Power Sources* **2010**, *195*, 2419−2430.
(6) Dunn, B.; Kamath, H.; Tarascon, J.-M. *Science* **2011**, *334*, 928−935.
(7) Quartarone, E.; Mustarelli, P. *Chem. Soc. Rev.* **2011**, *40*, 2525−2540.
(8) Robertson, A. D.; West, A. R.; Ritchie, A .G. *Solid State Ionics* **1997**, *104*, 1−11.
(9) Xu, X.; Bates, J. B.; Jellison, G. E.; Hart, F. X. *J. Electrochem. Soc.* **1997**, *144*, 524−532.
(10) Kawai, H.; Kuwano, J. *J. Electrochem. Soc.* **1994**, *141*, L78−L79.
(11) Stramare, S.; Thangadurai, V.; Weppner, W. *Chem. Mater.* **2003**, *15*, 3974−3990.
(12) Awaka, J.; Kijima, N.; Hayakawa, H.; Akimoto, J. *J. Solid State Chem.* **2009**, *182*, 2046−2052.
(13) Cussen, E. J. *J. Mater. Chem.* **2010**, *20*, 5167−5173.
(14) Thangadurai, V.; Kaack, H.; Weppner, W. *J. Am. Ceram. Soc.* **2003**, *86*, 437−440.
(15) Thangadurai, V.; Weppner, W. *Adv. Funct. Mater.* **2005**, *15*, 107−112.
(16) Murugan, R.; Thangadurai, V.; Weppner, W. *J. Electrochem. Soc.* **2008**, *155*, A90−A101.
(17) O'Callaghan, M. P.; Cussen, E. J. *Solid State Sci.* **2008**, *10*, 390−395.
(18) (a) Ohta, S.; Kobayashi, T.; Asaoka, T. *J. Power Sources* **2011**, *196*, 3342−3345. (b) Murugan, R.; Thangadurai, V.; Weppner, W. *Angew. Chem., Int. Ed.* **2007**, *46*, 7778−7781. (c) Awaka, J.; Kijima, N.; Hayakawa, H.; Akimoto, J. *J. Solid State Chem.* **2009**, *182*, 2046−2052.
(19) Narayanan, S.; Thangadurai, V. *J. Power Sources* **2011**, *196*, 8085−8090.
(20) (a) Ramzy, A.; Thangadurai, V. *Appl. Mater. Interfaces* **2010**, *2*, 385−390. (b) Narayanan, S.; Epp, V.; Wilkening, M.; Thangadurai, V. *RSC Adv.* **2012**, *2*, 2553−2561.
(21) Adams, S.; Rao, R. P. *J. Mater. Chem.* **2012**, *22*, 1426−1434.
(22) (a) Cussen, E. J. *J. Mater. Chem.* **2010**, *20*, 5167−5173. (b) Cussen, E. J. *Chem. Commun.* **2006**, 412−413.
(23) Kuhn, A.; Narayanan, S.; Spencer, L.; Goward, G.; Thangadurai, V.; Wilkening, M. *Phys. Rev. B* **2011**, *83*, 094302−1−11.
(24) (a) Geiger, C. A.; Alekseev, E.; Lazic, B.; Fisch, M.; Armbruster, T.; Langner, R.; Fechtelkord, M.; Kim, N.; Pettke, T.; Weppner, W. *Inorg. Chem.* **2011**, *50*, 1089−1097. (b) Buschmann, H.; Dolle, J.; Berendts, S.; Kuhn, A.; Bottke, P.; Wilkening, M.; Heitjans, P.; Senyshyn, A.; Ehrenberg, H.; Lotnyk, A.; et al. *Phys. Chem. Chem. Phys.* **2011**, *13*, 19378−19392.
(25) Phipps, A. M.; Hume, D. N. *J. Chem. Educ.* **1968**, *25*, 664.
(26) Larson, A. C.; Von Dreele, R. B. *General Structure Analysis System (GSAS)*, Los Alamos National Laboratory Report LAUR, 1994, 86−748.
(27) Toby, B. H. *J. Appl. Crystallogr.* **2001**, *34*, 210−213.
(28) Xie, H.; Alonso, J. A.; Li, Y.; Fernández-Díaz, M. T; Goodenough, J. B. *Chem. Mater.* **2011**, *23*, 3587−3589.
(29) Shannon, R. D. *Acta Crystallogr.* **1976**, *A32*, 751−767.
(30) O'Callaghan, M. P.; Cussen, E. J. *Chem. Commun.* **2007**, 2048−2050.
(31) O'Callaghan, M. P.; Lynham, D. R.; Cussen, E. J.; Chen, G. Z. *Chem. Mater.* **2006**, *18*, 4681−4689.
(32) (a) O'Dell, L. A.; Savin, S. L. P.; Chadwick, A. V.; Smith, M. E. *Solid State Nucl. Magn. Reson.* **2007**, *31*, 169−173. (b) Koch, B.; Vogel, M. *Solid State Nucl. Magn. Reson.* **2008**, *34*, 37−43.
(33) (a) Wullen, L. V.; Echelmeyer, T.; Meyer, H. W.; Wilmer, D. *Phys. Chem. Chem. Phys.* **2007**, *9*, 3298−3303. (b) Gupta, A.; Murugan, R.; Paranthaman, M. P.; Bi, Z.; Bridges, C. A.; Nakanishi, M.; Sokolov, A. P.; Han, K. S.; Hagaman, E. W.; Xie, H.; et al. *J. Power Sources* **2012**, *209*, 184−188.
(34) Wang, W.; Wang, X. -P.; Gao, Y. -X.; Yang, J. -F.; Fang, Q. -F. *Front. Mater. Sci. China* **2010**, *4*, 189−194.
(35) Li, Q.; Thangadurai, V. *Fuel Cells* **2009**, *9*, 684−698.
(36) Irvine, J. T. S.; Sinclair, D. C.; West, A. R. *Adv. Mater.* **1990**, *2*, 132−138.
(37) Almond, D. P.; Duncan, G. K.; West, A. R. *Solid State Ionics* **1983**, *8*, 159−164.
(38) París, M. A.; Sanz, J. *Chem. Mater.* **2000**, *12*, 1694−1701.
(39) Jonscher, A. K. *Nature* **1977**, *267*, 673−679.
(40) Jonscher, A. K. *Dielectric Relaxation in Solids*; Chelsea Dielectric Press: London, 1983.
(41) Hodge, I. M.; Ingram, M. D.; West, A. R. *J. Am. Ceram. Soc.* **1976**, *59*, 360−366.
(42) Thangadurai, V.; Adams, S.; Weppner, W. *Chem. Mater.* **2004**, *16*, 2998−3006.
(43) (a) Nyman, M.; Alam, T. M.; McIntyre, S. K.; Bleier, G. C.; Ingersoll, D. *Chem. Mater.* **2010**, *22*, 5401−5410. (b) Truong, L.; Thangadurai, V. *Chem. Mater.* **2011**, *23*, 3970−3977.